

PRE-UNIVERSITY COURSE (PUC) 2-YEAR CURRICULUM FRAMEWORK

Foundational Sciences, Mathematics, English Communication & Digital Literacy (P1 & P2)

Ref: RGUKT/PUC/CURR-2024

Effective: PUC AY 2024-26

Authority: PUC Academic Board, RGUKT-AP

1. Objectives of the Pre-University Foundation

The 2-year Pre-University Course (equivalent to Intermediate in Andhra Pradesh) bridges high school education with university engineering studies. The syllabus imparts deep conceptual mastery in basic sciences without premature compartmentalization.

- Mathematics Core (M1, M2, M3, M4):** Algebra, Trigonometry, Coordinate Geometry, Differential & Integral Calculus, Vectors, Probability.
- Physics Core & Laboratory:** Mechanics, Waves & Oscillations, Thermodynamics, Electromagnetism, Modern Physics, Optics.
- Chemistry Core & Laboratory:** Physical Chemistry, Inorganic Chemistry, Organic Reaction Mechanisms, Environmental Chemistry.
- Information Technology (IT):** Computer Architecture, Python Programming, Web Technologies, Database Basics, Algorithm Design.
- English & Soft Skills (~30% time):** Phonetics, Vocabulary Building, Technical Report Writing, Group Discussions, Classical Arts.